Capstone Project

Guideline Document

AI Training for Engineers

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Capstone Project Guidelines

Introduction

This document is designed to help learners gain a clear understanding of the Capstone Project, its purpose, and the expectations associated with it. The Capstone Project serves as a culminating exercise, allowing learners to apply the knowledge and skills acquired throughout the program in a practical and meaningful way.

Learners are encouraged to engage actively with their respective Training Advisors to get the capstone project requirements, understand evaluation criteria, and align on expected deliverables. Regular discussions with the advisor will help ensure that the project progresses in the right direction and meets the defined objectives.

By the end of the Capstone Project, learners should be able to demonstrate their problem-solving abilities, technical expertise, and overall readiness to apply their learning in real-world scenarios.

Building AI Solutions (Open-Source Only)

Design and implement an AI-based solution using **open-source models and tools only**, demonstrating strong understanding of model selection, evaluation, and responsible AI practices.

1. Model Usage Policy

✅ Allowed

- ✅ Open-source models only (Hugging Face / GGUF / etc.)
- ✅ Local inference setups

Restrictions

- ❌ Do NOT use proprietary or paid APIs:
  - OpenAI (GPT APIs)
  - Google Gemini
  - Azure OpenAI
- ❌ Do NOT rely on hosted inference endpoints from closed vendors

2. Recommended Tools for Running Models

- Ollama (Preferred for local LLM usage) **or** LM Studio (UI-based experimentation)

Supported models (examples):

- Llama 3
- Mistral
- Phi-3
- Gemma

3. Model Selection & Justification

Evaluate and justify model selection with the below parameters.

Criteria	Description
Accuracy	Quality of outputs
Latency	Response time
Cost	Compute/memory usage
Size	Model parameters
Use case fit	Domain suitability

Note : Good to have Compare at least 2 models

Show more lines

- Provide **quantitative justification** for selecting the final model.

4. Dataset Guidelines

🚫 Strict Restrictions

- ❌ Do NOT use:
  - Company internal data
  - Confidential datasets
  - Customer data
  - Proprietary documents

✅ Allowed Sources

- Public datasets (Kaggle, Hugging Face Datasets)

- Synthetic/generated data
- Open government datasets

5. Prompt Engineering / Optimization

- Try multiple prompt strategies:
    - Zero-shot
    - Few-shot
    - Chain-of-thought
- Document:
    - Prompt variations
    - Observed improvements

6. Documentation Deliverables

Report must include:

1. Problem Statement
2. Dataset Source
3. Model Comparison
4. Selection Justification (with scoring)
5. Demo Results
6. Limitations & Future Work
7. Documentation of Prompts used

7. Demo Requirements

- Live demo OR recorded walkthrough
- Show:
    - Input → Output
    - Model behavior
    - Edge cases

8. Evaluation Rubrics

S.No.	Parameters	Points
1	Functional Completeness	30
2	RAG Design & Retrieval Quailty	25
3	Prompt Engineering Quality	20
4	Python Code Quality & Structure	20
5	AI- Assisted Development Usage	15
6	Testing & Reliability	15
7	Engineering Judgment & Trade-offs (15 points)	15
8	Documentation & Communication (10 points)	10
Total Points		150

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